

HW1: Twitter in a Relational Database

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Hardware Configuration

I have a MacBook Pro with Apple M1 chip which has 8 cores, 8 GB unified memory, 3.2 GHz, 228 GB SSD (Apple APFS), running macOS 13 Ventura.

Software Stack

MySQL 9.6.0 (installed via Homebrew), Python 3.12 (Anaconda), mysql-connector-python for database connectivity, and standard libraries (pandas, csv, time, random, os) and for my IDE I used VsCode.

Database Design

Two tables: tweet (tweet_id, user_id, tweet_ts, tweet_text, and follows. I added 3 secondary indexes: on follows.follower_id, on tweet.user_id, and a composite index on tweet(user_id, tweet_ts DESC) to speed up the ORDER BY in timeline queries.

Performance Results

postTweet: 2487 API calls/sec

getTimeline: 314 API calls/sec

Analysis

Twitter handles 6000–10000 tweets/sec and 200000–300000 timeline refreshes/sec globally. Compared to my MySQL implementation, I achieved only 2487 inserts/sec and 314 timeline retrievals/sec which is extremely but expectantly far short of production demands, especially for reads.

There are several factors that could have impacted results. One is that each insert writes to the disk and updates the 3 secondary indexes and this adds a per-insert overhead. Adding onto this the getTimeline query needs a JOIN between follows and tweet, an ORDER BY on timestamp, and conversion of the result set into Tweet objects. Even while using indexes this is still pretty expensive because the tweet table is growing to a million rows. Furthermore, the Python mysql-connector also adds serialization overhead compared to compiled languages. One last thing that I learned is that in my case, my python program and MySQL are both running on the same laptop, so they would be fighting over the same CPU and RAM. However in actual production environments, like at Twitter, the application and database would run on separate servers. These limitations can give reason to switch to Redis in HW2 since Redis stores data on RAM instead of on disk making reads an incredible amount faster.

AI Usage

I used Claude to help me debug my MySQL installation and terminal/path issues as well as creating an outline for the report/analysis. I also used claude to figure out what commands to put in my terminal to figure out the hardware configuration of my laptop